1.1. Design and Analysis of a Butterworth IIR Low-Pass Filter Using MATLAB.

```
clc;
sponse
figure;
freqz(b,a);
title('Magnitude and Phase Response');

% Objective 3: Impulse Response
figure;
clear;
close all;

% Butterworth IIR Low Pass Filter
Fs = 10000;
Fp = 1000;
Fst = 1500;

[N,Wn] = buttord(Fp/(Fs/2), Fst/(Fs/2), 1, 60);
[b,a] = butter(N,Wn);

% Objective 2: Magnitude and Phase Re
impz(b,a,50);
title('Impulse Response');
% Step Response
figure;
stepz(b,a);
title('Step Response');

% Objective 4: Pole-Zero Plot
figure;
zplane(b,a);
title('Pole-Zero Plot');

fprintf('Minimum Filter Order (N) = %d\n',N);
fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```

Output

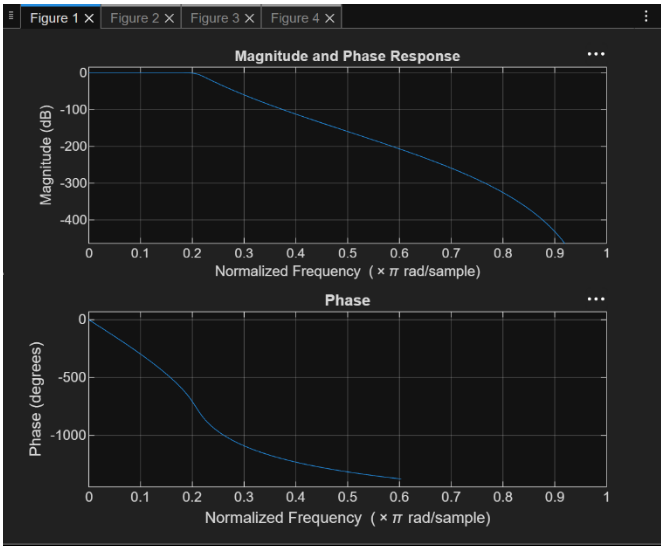
Figure 1 Figure 2 Figure 3 Figure 4
Magnitude and Phase Response
Magnitude (dB)
Normalized Frequency (×π rad/sample)
Phase
Phase (degrees)
Normalized Frequency (×π rad/sample)

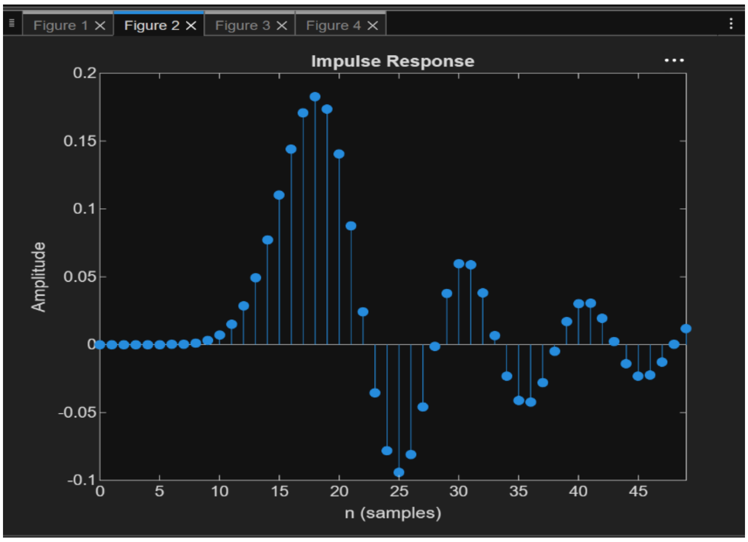
Figure 1 Figure 2 Figure 3 Figure 4
Impulse Response
Amplitude
n (samples)

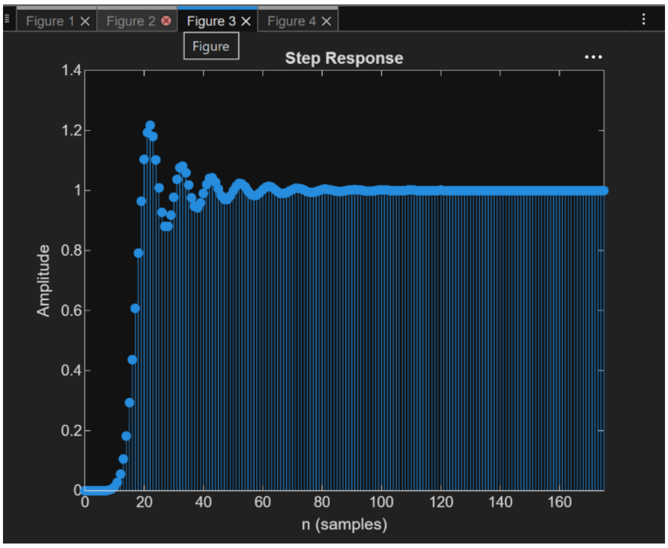
Figure 1 Figure 2 Figure 3 Figure 4
Figure
Step Response
Amplitude
n (samples)

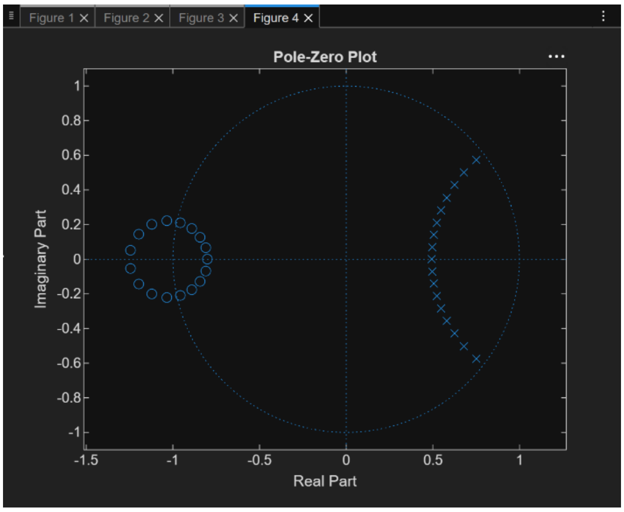
Figure 1 Figure 2 Figure 3 Figure 4
Pole-Zero Plot
Imaginary Part
Real Part

1.2. Design and Analysis of a Butterworth IIR High-Pass Filter Using MATLAB.

```
clc;
clear;
close all;


% Filter Specifications
Fs  = 10000;      % Sampling Frequency (Hz)
Fp  = 3000;       % Passband Frequency (Hz)
Fst = 2000;       % Stopband Frequency (Hz)
Rp  = 1;          % Passband Ripple (dB)
Rs  = 60;         % Stopband Attenuation (dB)

[N, Wn] = buttord(Fp/(Fs/2), Fst/(Fs/2), Rp, Rs);

% Design Butterworth High-Pass Filter
[b, a] = butter(N, Wn, 'high');


figure;
freqz(b, a, 1024, Fs);
title('Magnitude and Phase Response');

figure;
[h, n] = impz(b, a, 50);
stem(n, h, 'filled');
grid on;
xlabel('Samples');
ylabel('Amplitude');
title('Impulse Response');

figure;
x = ones(1,50);           % Unit Step Input
y = filter(b, a, x);      % Step Response

stem(0:49, y, 'filled');
grid on;
xlabel('Samples');
ylabel('Amplitude');
title('Step Response');
```

```
figure;
grpdelay(b, a, 1024, Fs);
grid on;
title('Group Delay');

% Pole-Zero Plot
figure;
zplane(b, a);
grid on;
title('Pole-Zero Plot');

% Display Filter Details

fprintf('Minimum Filter Order (N) = %d\n', N);
fprintf('Normalized Cutoff Frequency (Wn) = %.4f\n', Wn);
```